Curs 11
Recuperare

14.XII.2012

Metode iterative de rezolvare a sistemelor

$\|A\|_1 \qquad \|A\|_\infty$

$x=(x_1,\dots,x_n)^T \qquad \|x\|_\infty=\max_{i=\overline{1,n}}|x_i|$

$\|A\|_\infty=? \quad A=[a_{i,j}]$

Fie $x\in\mathbb{C}^n$: $\|x\|_\infty=1$. $\Rightarrow$ cea mai mare componentă e 1

$\|Ax\|_\infty$.

$Ax=(y_1,y_2,\dots,y_n)^T$

$y_i=\sum_{j=1}^n a_{i,j}x_j$

$$\|Ax\|_\infty=\max_{j=\overline{1,n}}\left(\left|\sum_{j=1}^n a_{i,j}x_j\right|\right)\le\max_{i=\overline{1,n}}\left(\sum_{j=1}^n|a_{i,j}||x_j|\right)\le\underbrace{\|x\|_\infty}_{1}\max_{i=\overline{1,n}}\sum_{j=1}^n|a_{i,j}|$$

$\|Ax\|_\infty\le\max_{i=\overline{1,n}}\sum_{j=1}^n|a_{i,j}|$

trecem la supremum

$\Rightarrow \|A\|_\infty\le c,\quad c=\max_{i=\overline{1,n}}\left(\sum_{j=1}^n|a_{i,j}|\right)\ \underline{\underline{\exists k\in\{1,\dots,n\}}}\ \sum_{j=1}^n|a_{k,j}|$

arătăm ca $\|A\|_\infty=c$. (găsesc un vector care are norma în ∞, 1 care satisface egalitatea)

- construim vectorul

$x=(x_1,\dots,x_n)^T$, $x_j=\begin{cases}\frac{|a_{k,j}|}{a_{k,j}}, & a_{k,j}\neq0\\ 1, & |a_{k,j}|=0\end{cases}$

$|x_j|=1,\ \forall j\in\overline{1,n}$.

$\|x\|_\infty=1$.

- arătăm că satisface egal

$$\left|\sum_{j=1}^n a_{i,j}x_j\right|=\left|\sum_{j=1}^n a_{i,j}\frac{|a_{k,j}|}{a_{k,j}}\right|\le\sum_{j=1}^n|a_{i,j}|\cdot\frac{|a_{k,j}|}{|a_{k,j}|}\le\sum_{j=1}^n|a_{k,j}|$$

$i=k$

$$\left|\sum_{j=1}^n \cancel{a_{k,j}}\frac{|a_{k,j}|}{\cancel{a_{k,j}}}\right|=c. \quad\Rightarrow \boxed{\|A\|_\infty=c}$$

$\lim\limits_{k\to\infty} A^k = 0$ dacă $\exists$ o normă de matrice a.î. $\|A\| < 1$

$\{\lambda_1, \lambda_2, \dots, \lambda_n\} = \sigma(A)$ $\quad \exists P$ a.î. $P^{-1}AP = \Lambda$.

$$A = P\Lambda P^{-1}$$

$$A^k = P\Lambda^k P^{-1}$$

$$\|A^k\| \le \|P\| \cdot \underbrace{\|\Lambda^k\|}_{\searrow 0} \cdot \|P^{-1}\| \to 0$$

$\Rightarrow \lim\limits_{k\to\infty} A^k = 0.$

$$\Lambda = \overbrace{\operatorname{diag}(\lambda_1, \dots, \lambda_p)}^{D} + N$$

$$N^n = O_2$$

$$\Lambda^k = \sum_{i=0}^{k} \binom{k}{i} D^i N^{k-i} = \sum_{i=0}^{k} \binom{k}{i} D^{k-i} N^k \overset{\text{dacă } k>n}{=} \sum_{i=0}^{n} \binom{k}{i} \underset{\substack{\downarrow k\to\infty \\ 0}}{D^{k-i}} N^i \to O_n$$

$X = b + BX$

$Ax = b$.

not $L = \begin{pmatrix} 0 & \dots & & 0 \\ a_{21} & 0 & \dots & \\ a_{31} & a_{32} & 0 \dots & 0 \\ \dots & & & \\ a_{n1} & a_{n2} & \dots a_{n,n-1} & 0 \end{pmatrix}$, $U = \begin{pmatrix} 0 & a_{12} & \dots & a_{1n} \\ 0 & 0 & a_{23} \dots & a_{2n} \\ - & - & - & - \\ 0 & \dots & \dots & 0 \end{pmatrix}$, $D = \begin{pmatrix} a_{11} & & 0 \\ & a_{22} & \\ 0 & & \ddots\ a_{nn} \end{pmatrix}$

$A = L + D + U$

$Ax = b$

$(L + D + U)x = b \qquad$ Pp $a_{ii} \ne 0$

$Dx = b - (L+U)x \quad | D^{-1}$

$x = D^{-1}b - D^{-1}(L+U)x$

not $B = -D^{-1}(L+U)$

$\Rightarrow \underbrace{\{x = b' + Bx\}}$

$D^{-1} = \begin{pmatrix} \frac{1}{a_{11}} & \dots & & 0 \\ \vdots & \frac{1}{a_{22}} & & \vdots \\ \vdots & & \ddots & \vdots \\ 0 & \dots & & \frac{1}{a_{nn}} \end{pmatrix}$

$x = b + Bx$ (*), $B \in M_n(\mathbb{R})$

$x^{k+1} = b + Bx^k$ (**), $x^0 \in \mathbb{C}^n$

($k+1$ - indice)

[T] Condiția necesară și suficientă pt ca sistemul (*) să aibă soluție unică și șirul x^k construit prin relația (**) să conveargă către soluția sistemului este ca $\lim_{k\to\infty} B^k = 0$ (raza spectrală a lui $B < 1$)

(să $\exists$ o normă cu prop că $\|B\| < 1$)

Dem

$x = Bx$ (sist omogen)

$x^{k+1} = Bx^k$

$x^0 \in \mathbb{C}^m$ pct arbitrar

$x^1 = Bx^0$

$x^2 = Bx^1 = B^2x^0$

$\Rightarrow x^k = B^k x_0$ (B^k se citește aplicat lui x_0) $\Rightarrow \lim_{k\to\infty} x^k = 0$

$\lim_{k\to\infty} B^k = 0$

$\exists \|B\| < 1$

$\Rightarrow (I-B)^{-1} = \sum_{k=0}^{\infty} B^k$

$(I-B)x = b$

$x = (I-B)^{-1} b$ //

$x = b + Bx$

$x^k = b + Bx^{k-1}$

$x - x^k = B(x - x^{k-1})$

$\Rightarrow x - x^k = B^k(x - x^0)$

$\|x - x^k\| = \|B^k(x - x^0)\| \leq \|B^k\| \cdot \|x - x^0\|$

$$x^{k+1} = b + Bx^k$$
$$x^k = b + Bx^{k-1} \quad \Rightarrow \quad x^{k+1} - x^k = B(x^k - x^{k-1})$$

not $y^k = x^k - x^{k-1} \Longrightarrow y^{k+1} = By^k \Rightarrow y^{k+1} = B^k y^1$

$$x^{k+1} - x^k = B^k(x^1 - x^0)$$

$$\Rightarrow \|x^{k+1} - x^k\| \le \|B^k\| \cdot \|x^1 - x^0\|$$

$$\|x^{k+p} - x^k\| = \|(x^{k+p} - x^{k+p-1}) + (x^{k+p-1} - x^{k+p-2}) + \dots + (x^{k+1} - x^k)\|$$
$$\le \|x^{k+p} - x^{k+p-1}\| + \|x^{k+p-1} - x^{k+p-2}\| + \dots + \|x^{k+1} - x^k\|$$
$$\le (\|B^{k+p-1}\| + \|B^{k+p-2}\| + \dots + \|B^k\|)\|x^1 - x^0\|$$

T Dacă $\exists\, q \in (0,1)$ a.î. $\|B\| < q$ atunci $\|x - x^k\| \le \dfrac{q^k}{1-q}\|x^1 - x^0\|$

dem

$$\|x^{k+p} - x^k\| \le (q^{k+p-1} + q^{k+p-2} + \dots + q^k)\|x^1 - x^0\|$$
$$= \frac{q^k(1 - q^p)}{1-q}\|x^1 - x^0\|$$

$p \to \infty \Rightarrow \boxed{\|x - x^k\| \le \dfrac{q^k}{1-q}\|x^1 - x^0\|}$

Def Fie $A = [a_{ij}]$, $i,j = \overline{1,m}$, $a_{ij} \in \mathbb{C}$. Matricea A s.n. diagonal dominantă dacă $\sum_{\substack{j=1 \\ j \ne i}}^{n} |a_{ij}| \le |a_{ii}|$, $\forall i = \overline{1,n}$

$A = L + D + U$

$L = [a_{ij}]$, $a_{i,j} = 0$, $i \ge j$

$U = [a_{ij}]$, $a_{ij} = 0$, $1 \le i \le j$

pp $a_{ii} \ne 0$, $\forall i = \overline{1,m}$

$Ax = b$

$Dx = b - (L+U)x$

$x = D^{-1}b - D^{-1}(L+U)x$

$$x^{k+1} = D^{-1}b - D^{-1}(L+U)x^k$$

dacă $x^k = (x_1^k, x_2^k, \dots, x_n^k)^T$

$$\Rightarrow x_i^{k+1} = \frac{1}{a_{ii}}\Big(b_i - \sum_{\substack{j=1 \\ j\neq i}}^{n} a_{ij} x_j^k\Big), \quad i=\overline{1,n}$$

$$B = -D^{-1}(L+U)$$

$$\|B\|_\infty = \|D^{-1}(L+U)\|_\infty = \max_{i=\overline{1,n}} \sum_{j\neq i} \frac{|a_{ij}|}{a_{ii}} < 1$$ dacă matr. sist. diagonal dominantă

[T] Fie sistemul $Ax=b$. Dacă matricea A este o matrice diagonal dominantă, atunci șirul $x^{k+1} = D^{-1}b - D^{-1}(L+U)x^k$ converge către unica soluție a sistemului.

Metoda lui Gauss - Seidel

$Ax=b$

$A = L+D+U$ $\qquad a_{ii}\neq 0, \forall i=\overline{1,n}$

$$(L+D)x + Ux = b$$

$$x = (L+D)^{-1}b - (L+D)^{-1}Ux$$

$$B = -(L+D)^{-1}U$$

$$x^{k+1} = (L+D)^{-1}b - (L+D)^{-1}Ux^k$$

$$(L+D)x^{k+1} = b - Ux^k$$

$$Dx^{k+1} = b - Lx^{k+1} - Ux^k$$

$$x^{k+1} = D^{-1}b - D^{-1}Lx^{k+1} - D^{-1}Ux^k$$

$$x_i^{k+1} = \frac{1}{a_{ii}}\Big(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{k+1} - \sum_{j=i+1}^{n} a_{ij} x_j^k\Big), \quad i=\overline{1,n}$$

$\boxed{T}$ Dacă matricea A îndeplinește condițiile:
$\sum_{\substack{j=1 \\ i\neq j}}^{n} |a_{ij}| \le |a_{ii}|$ și $\sum_{j=i+1}^{n} |a_{ij}| < |a_{ii}|$, $i=\overline{1,n}$ atunci
metoda lui Gauss-Seidel este o metodă convergentă.

dem

$B = -(L+D)^{-1}U \qquad \|B\|_\infty < 1.$

$x \in \mathbb{C}^n : \|x\|_\infty = 1.$

$Bx = -(L+D)^{-1}Ux$

$y = -(L+D)^{-1}Ux \mid (L+D)$

$(L+D)y = -Ux$

$Dy = -Ux - Ly$

$y = -D^{-1}(Ux+Ly)$

$$y_i = -\frac{1}{a_{ii}}\left(\sum_{j=i+1}^{n} a_{ij}x_j + \sum_{j=1}^{i-1} a_{ij}y_j\right)$$

$$y_1 = -\frac{1}{a_{11}}\sum_{j=2}^{n} a_{1j}$$

$$|y_1| \le \frac{1}{|a_{11}|}\sum_{j=2}^{n} |a_{1j}| < 1$$

Pp că $|y_{i-1}| < 1$

dem că $|y_i| < 1$.

$$|y_i| \le \frac{1}{|a_{ii}|}\left(\sum_{j=i+1}^{n} |a_{ij}||x_j| + \sum_{j=1}^{i-1} |a_{ij}|\right) \le \frac{1}{|a_{ii}|}\sum_{\substack{j=1 \\ j\neq i}}^{n} |a_{ij}| < 1$$

$\Rightarrow \|B\|_\infty < 1$